- **Encryption and Decryption:** OpenSSL empowers the encryption and decryption of data, particularly crucial for securing sensitive information, such as user credentials, personal data, and financial transactions. In the Laravel framework, OpenSSL is harnessed to establish secure communication through HTTPS (HTTP Secure), encrypting data during transmission.
- **SSL/TLS Protocols**: OpenSSL lends support to various SSL (Secure Sockets Layer) and TLS (Transport Layer Security) protocols, vital for securing communication between a web server and a client (such as a web browser). These protocols guarantee that the data exchanged between the client and the server remains encrypted, shielding it from unauthorized access.
- **Digital Signatures**: Offering functions for creating and verifying digital signatures, OpenSSL plays a key role in ensuring the integrity and authenticity of data. In Laravel, digital signatures can be employed to authenticate data received from clients or other servers.
- **Certificate Handling**: SSL/TLS certificates constitute a critical component of secure web communication. OpenSSL simplifies the generation, management, and verification of SSL/TLS certificates. Laravel applications frequently leverage OpenSSL for SSL certificate management, establishing secure connections.
- **Hashing and Message Digests**: OpenSSL incorporates cryptographic hash functions that Laravel can utilize for tasks such as hashing passwords or generating message digests. Hashing is pivotal for securely storing passwords and ensuring data integrity.

OpenSSL equips developers with indispensable tools for encryption, decryption, digital signatures, and certificate management, bolstering the overall security of the application and its interactions with users and other services.

To utilize OpenSSL functions like **`openssl_pkey_new`**, you must first build an **`openssl.cnf`** configuration file and tell Vapor to load it using the **`OPENSSL_CONF`** environment variable. For example, this environment variable instructs Vapor to import an **`openssl.cnf`** file from your project's root:

```
OPENSSL_CONF="/var/task/openssl.cnf"
```

Let's look at the sample **`openssl.cnf`** file: